

Q No. 1
Question

Calculate mass of 4.5 gm atoms of Magnesium (Mg = 24)

- A) 84 gm
- B) 108 gm
- C) 24 gm
- D) 156 gm

Correct Answer

108 gm

Solution

1 gm atom of Mg = 24 gm

4.5 gm atom of Mg = $24 \times 4.5 = 108$ gm

Q No. 2
Question

Calculate the mass of 1.5 gm of molecules of H_2SO_4 (S = 32)

- A) 98 gm
- B) 148 gm
- C) 147 gm
- D) 100 gm

Correct Answer

147 gm

Solution

1 gm molecule of $\text{H}_2\text{SO}_4 \Rightarrow 2 \times 1 + 32 + 4 \times 16$

$\Rightarrow 98$ gm

1.5 gm of molecule of $\text{H}_2\text{SO}_4 \Rightarrow 98 \times 1.5$

$\Rightarrow 147$ gm

Q No. 3
Question

B has two isotopes ^{10}B (19%) and ^{11}B (81%). The atomic mass of B is

- A) 10.81
- B) 11.5
- C) 11
- D) 10.5

Correct Answer

10.81

Solution

The handwritten solution shows a tree diagram for element B branching into two isotopes: B^{10} and B^{11} . Below the diagram, the percentages are listed as 19% for B^{10} and 81% for B^{11} . The calculation for the Atomic Mass (AAM) is shown as follows:

$$\begin{aligned} \text{AAM} &= \frac{10 \times 19 + 11 \times 81}{100} \\ &= \frac{1081}{100} \\ &= 10.81 \end{aligned}$$

Q No. 4
Question

3 g of a hydrocarbon on combustion with 11.2 grams of oxygen produces 8.8 g of CO_2 and 5.4 g H_2O . The data illustrates the law of -

- A) Conservation of mass
- B) Multiple proportion
- C) Constant proportions
- D) Reciprocal Proportions

Correct Answer

Conservation of mass

Solution

NA

Q No. 5
Question

6 g of carbon combines with 32 g of sulphur to form CS_2 , 12 g of C also combine with 32 g oxygen to form CO_2 · 10 g of sulphur combines with 10 g of oxygen to form Sulphur dioxide. Which law is illustrated by this?

- A) Law of multiple proportions
- B) Law of constant composition
- C) Law of reciprocal proportions
- D) Gay Lussac's law

Correct Answer

Law of reciprocal proportions

Solution

NA

Q No. 6
Question

The % of copper and oxygen in samples of CuO obtained by different methods were found to be the same. This proves the law of:

- A) Constant Proportion
- B) Reciprocal Proportion
- C) Multiple Proportion
- D) Conservation of mass.

Correct Answer

Constant Proportion

Solution

According to law of constant proportion

Q No. 7
Question

Two compound AB_2 and AB in which fixed mass of B combine with different masses of A represent-

- A) Law of conservation of mass
- B) Law of constant composition
- C) Law of multiple proportion
- D) Law of reciprocal proportion

Correct Answer

Law of multiple proportion

Solution

NA

Q No. 8

Question

What is the molar mass in gram of SO_4^{2-}

Correct Answer

96

Solution



$$32 + 4 \times 16 = 96$$